COMPUTER VISION

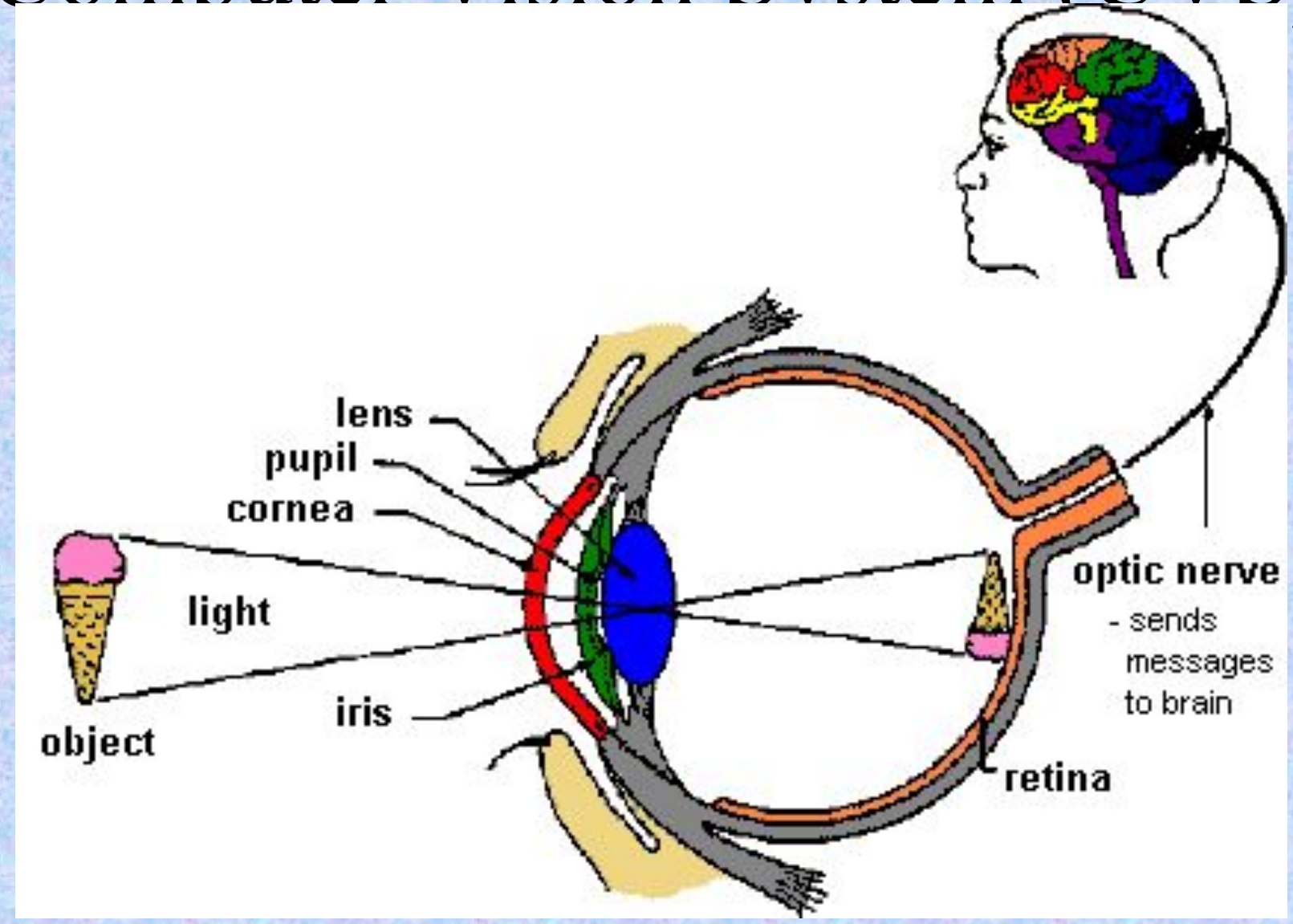
Reference Material:
//www.cse.iitm.ac.in/~vplab/computer_vision.html

1


jerel rowan baker

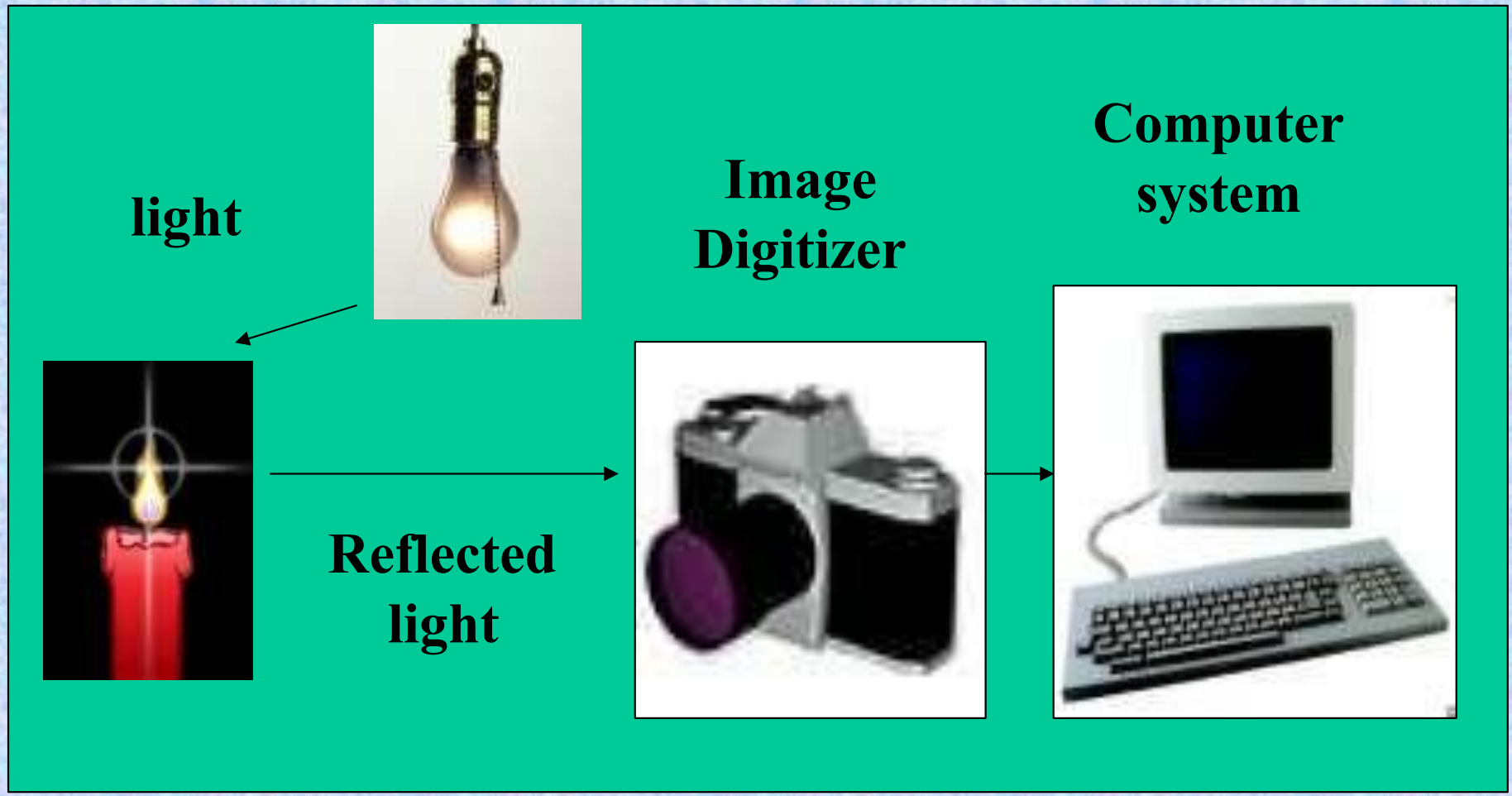


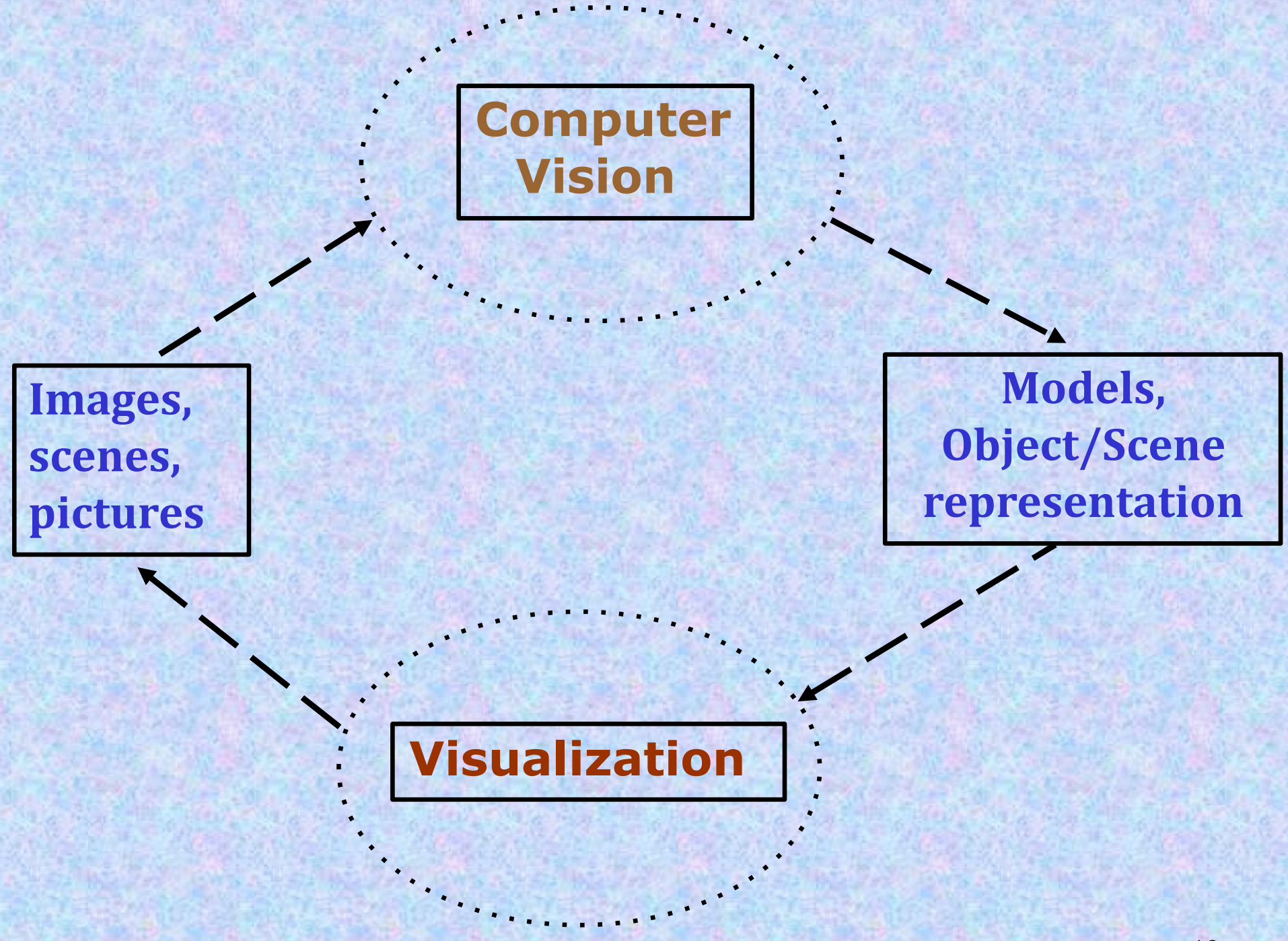
Human Vision System (HVS) Vs. Computer Vision System (CVS)



The Optics of the eye

A computer Vision System (CVS)



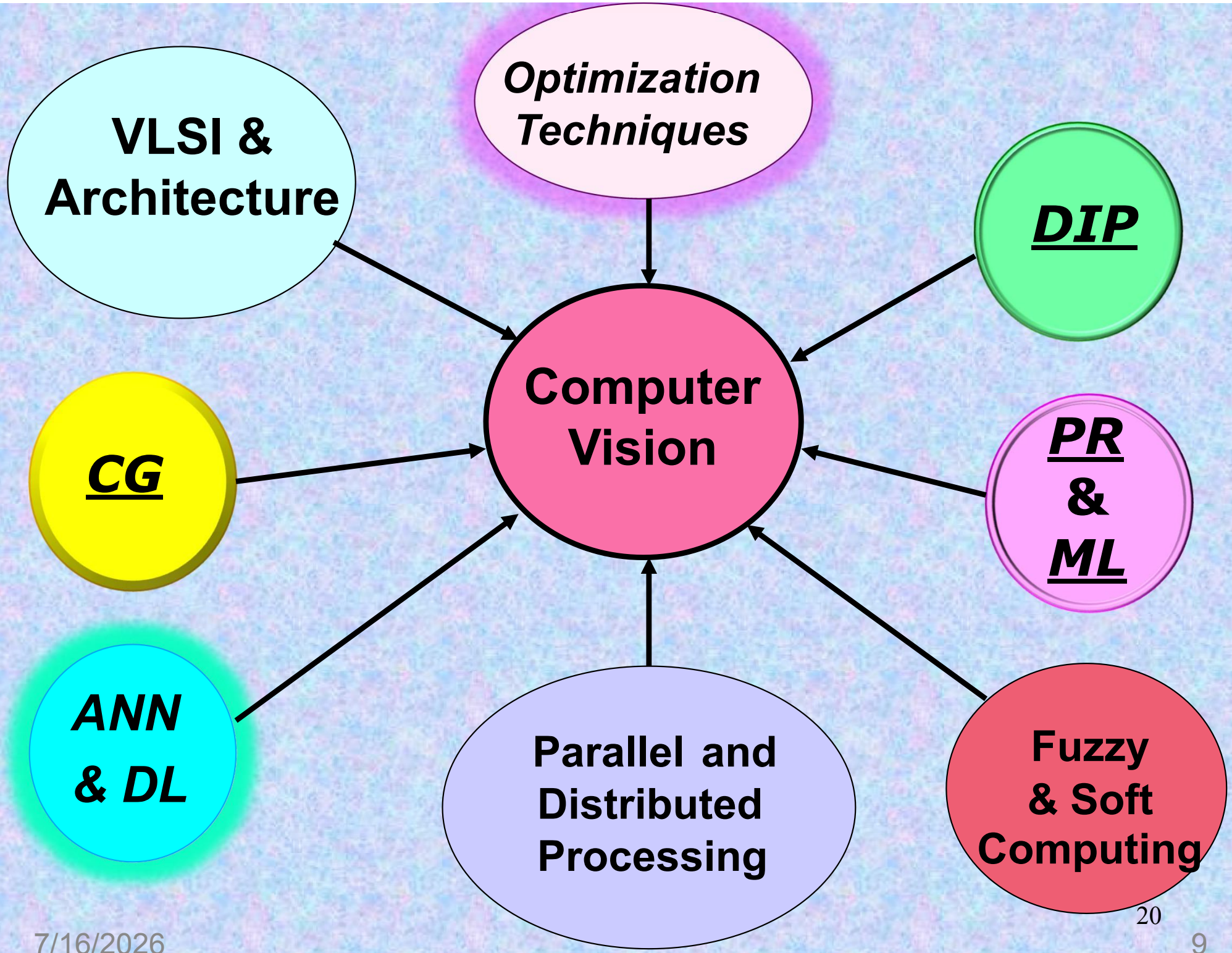

Computer Vision
Models, Object/Scene representation
Visualization
Images, scenes, pictures

Computer Vision **is an area of work, which is a combination of concepts, techniques and ideas from Digital Image Processing, Pattern Recognition, Artificial Intelligence and Computer Graphics.**

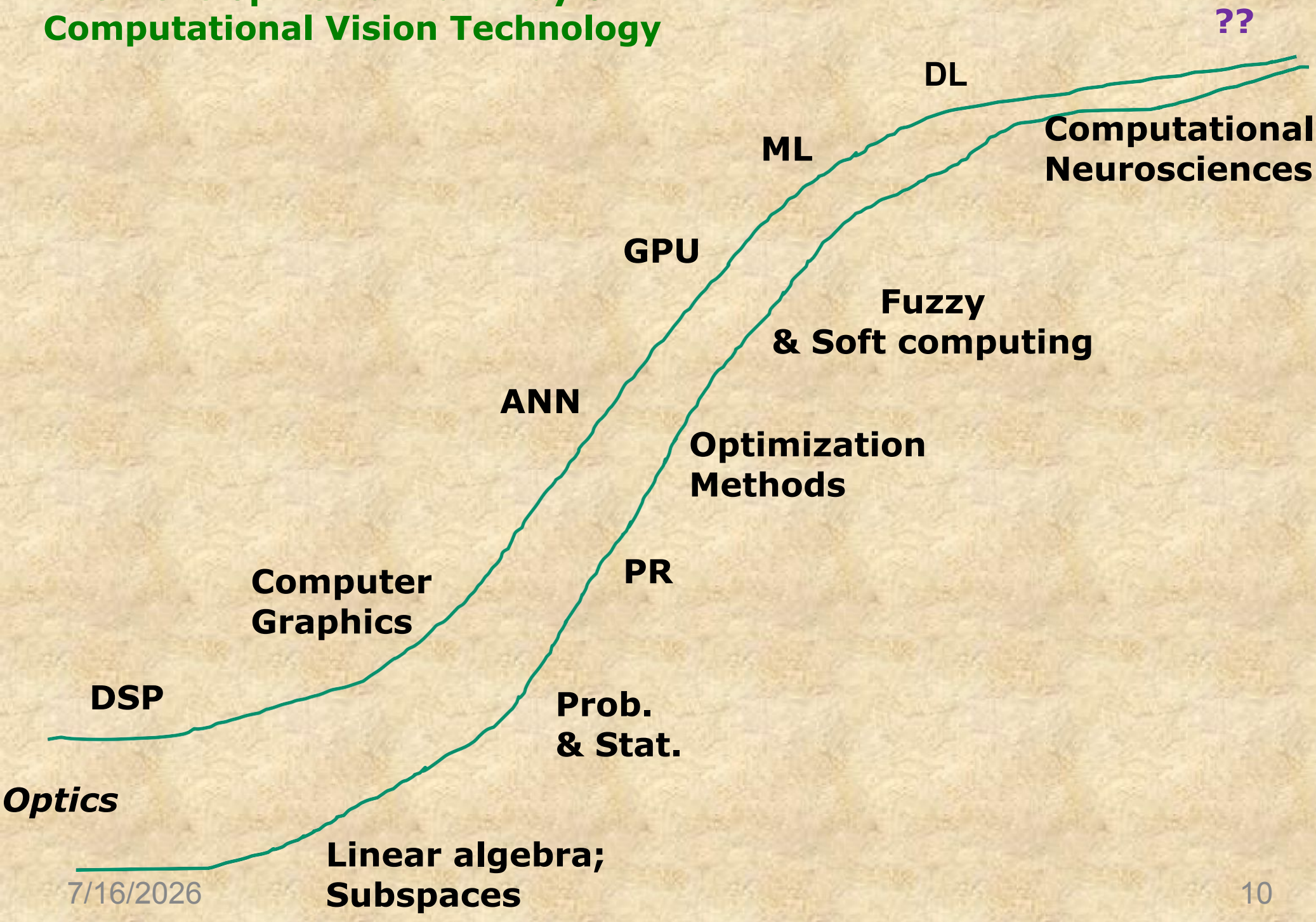
Majority of the tasks in the fields of Digital Image Processing or Computer Vision deals with the process of understanding or deriving the scene information or description, from the input scene (digital image/s). The methods used to solve a problem in digital image processing depends on the application domain and nature of data being analyzed.

Analysis of two-dimensional pictures are generally not applicable of processing three-dimensional scenes, and vice-versa. The choice of processing, techniques and methods and 'features' to be used for a particular application is made after some amount of trial and error, and hence experience in handling images is crucial in most of these cases.

For example, analysis of remote sensed or satellite imagery involves techniques based on classification or analysis of texture imagery. These techniques are not useful for analyzing optical images of indoor or outdoor scenes


VLSI & Architecture
Optimization Techniques
DIP
Computer Vision
CG
PR & ML
ANN & DL
Parallel and Distributed Processing
Fuzzy & Soft Computing

The Developmental Pathway of Computational Vision Technology



Digital Image processing **is in many cases concerned with taking one array of pixels as input and producing another array of pixels as output which in some way represents an improvement to the original array.**

Purpose:

1. **Improvement of Pictorial Information**
   - **improve the contrast of the image,**
   - **remove noise,**
   - **remove blurring caused by movement of the camera during image acquisition,**
   - **it may correct for geometrical distortions caused by the lens.**

2. **Automatic Machine perception (termed Computer Vision, Pattern Recognition or Visual Perception)for intelligent interpretation of scenes or pictures.**

Elements of a Digital Image Processing

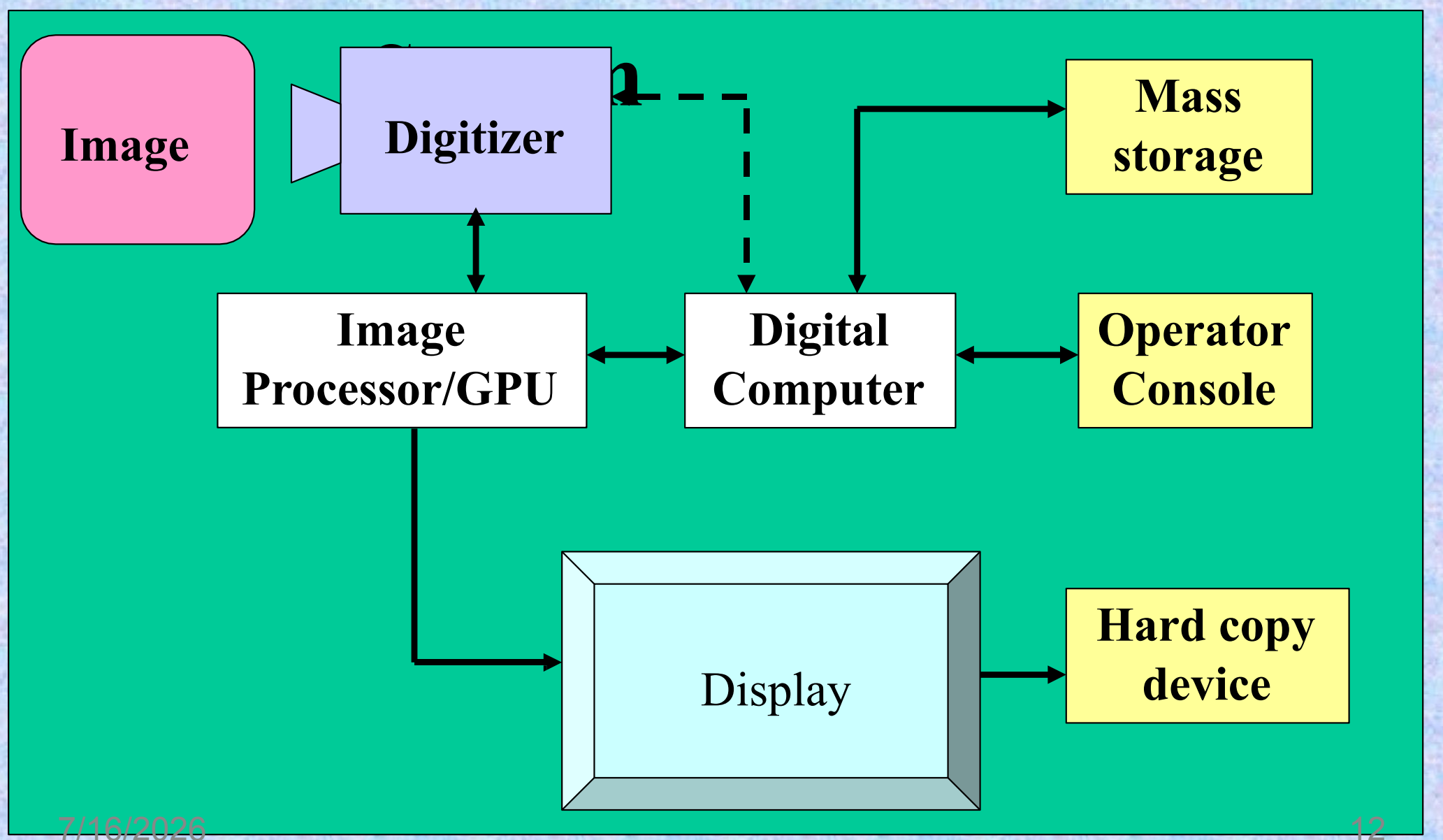


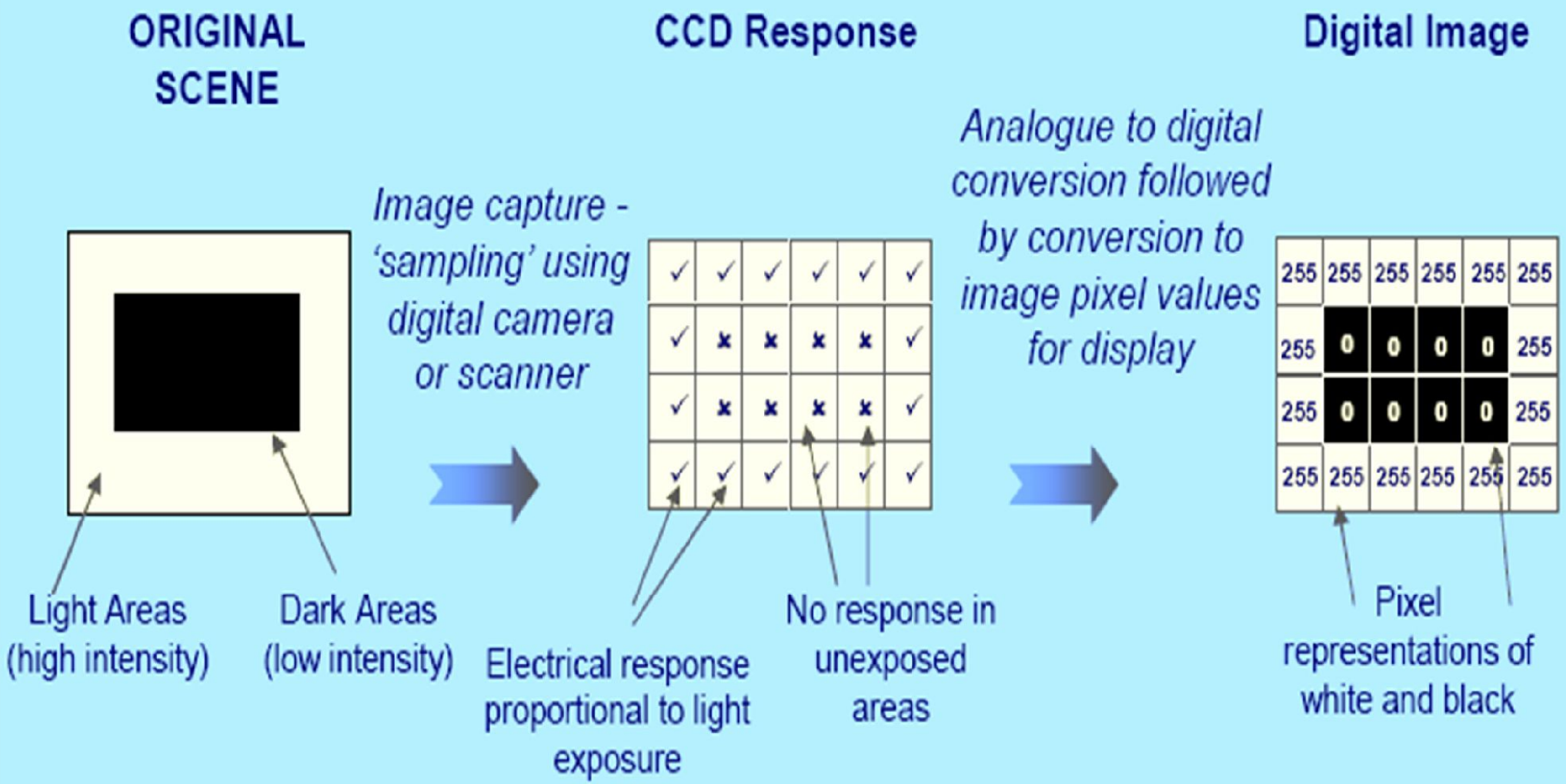
Image processors:

Consists of set of hardware modules that perform 4 basic functions:

- Image acquisition: frame grabber
- Storage: frame buffer
- Low-level processing: specialized hardware device designed to perform Arithmetic Logic operations on pixels in parallel
- Display: read from image memory (frame buffer) and convert to analog video signal

- **Digitizers:** Converts image into numerical representation suitable for input to a digital computer
- **Digital Computers:** Interfaced with the image processor to provide versatility and ease of programming.
- **Storage Devices:** For bulk storage. e.g:- Magnetic disks, magnetic tapes, optical disks
- **Display and Recording devices :** Monochrome and Color Television monitors, CRT, Laser printers, heat-sensitive paper devices, and ink spray systems

Image acquisition using a CCD camera



Resolution standards: HDMI - 1024*768; 10K UHD - ??

A digital Image

Image is an array of integers: $f(x,y) \varepsilon \{0,1,\ldots,I_{max}-1\}$,
where, $x,y \varepsilon \{0,1,\ldots,N-1\}$

- N is the resolution of the image and I_{max} is the level of discretized brightness value
- Larger the value of N, more is the clarity of the picture (larger resolution), but more data to be analyzed in the image
- If the image is a gray-level (8-bit per pixel - termed raw, gray) image, then it requires N^2 Bytes for storage
- If the image is color - RGB, each pixel requires 3 Bytes of storage space.

Image Size (resolution)	Storage space required	
	Raw - Gray	Color (RGB)
64*64	4K	12K
256*256	64K	192K
512*512	256K	768K

2048×1536 = 3.1 megapixels 9 MB for RGB

A digital image is a two-dimensional (3-D image is called range data) array of intensity values, f(x, y), which represents 2-D intensity function discretized both in spatial coordinates (spatial sampling) and brightness (quantization) values. The elements of such an array are called pixels (picture elements).

The storage requirement for an image depends on the spatial resolution and number of bits necessary for pixel quantization.

The processing of an image depends on the application domain and the methodology used to solve a problem. There exists four broad categories of tasks in digital image processing:

(i) Reconstruction

(ii)Segmentation,

(iii) Recognition and

(iv) motion.

Segmentation **deals with the process of fragmenting the image into homogeneous meaningful parts, regions or sub-images. Segmentation is generally based on the analysis of the histogram fo the images using gray level values as features. Other features used are edges or lines, colors and textures.**

Recognition **deals with identification or classification of objects in an image for the purpose of interpretation or identification. Recognition is based on models, which represent an object. A system is trained (using HMM, GMM, ANN etc.) to learn or store the models, based on training samples. The test data is then matched with all such models to identify the object with a certain measure of confidence.**

Reconstruction (Depth & 3D) - **3D reconstruction and depth estimation are critical techniques in computer vision that enable machines to interpret and model the three-dimensional structure of scenes from 2D images or video. 3D reconstruction involves creating a 3D model of an object or environment by integrating multiple 2D images, often using methods like structure-from-motion (SfM), multi-view stereo, or point cloud generation; leveraging geometric and photometric cues to reconstruct spatial geometry.**

Depth estimation, a closely related task, focuses on inferring the distance of objects or surfaces from a camera, typically using monocular cues (e.g., shading, texture gradients) or stereo vision, where disparity between images from multiple viewpoints is used. (Src: GROK)

Motion analysis (or dynamic scene analysis) **involves techniques for the purpose of tracking and estimation of the path of movement of object/s from a sequence of frames (digital video).**

Methods for dynamic scene analysis are based on (i) tracking, (ii) obtaining correspondence between frames and then (iii) estimating the motion parameters and (iv) structure of moving objects. Typical methods for analysis are based on optical flow, iterative Kalman filter and Newton/Euler's equations of dynamics.

There are generally three main categories of tasks involved in a complete computer vision system. They are:

- ***Low level processing*: Involves image processing tasks in which the quality of the image is improved for the benefit of human observers and higher level routines to perform better.**
- ***Intermediate level processing*: Involves the processes of feature extraction and pattern detection tasks. The algorithms used here are chosen and tuned in a manner as may be required to assist the final tasks of high level vision.**

- ***High level vision***: **Involves autonomous interpretation of scenes for pattern classification, recognition and identification of objects in the scenes as well as any other information required for human understanding.**

A top down approach, rather than a bottom-up approach is used in the design of these systems in many applications. The methods used to solve a problem in digital image processing depends on the applicat$_{3}$i$_{0}$on domain and nature of data being analyzed.

1. **Different fields of applications include:**

   Character Recognition,

   Document processing,

   Commercial (signature & seal verification) application,

   Biometry and Forensic (authentication: recognition and verification of persons using face, palm & fingerprint),

   Pose and gesture identification,

   Automatic inspection of industrial products, Industrial process monitoring,

   Biomedical Engg. (Diagnosis and surgery),

   Military surveillance and target identification, **Navigation and mobility (for robots and unmanned vehicles - land, air and underwater),**

   Remote sensing (using satellite imagery), GIS

   Safety and security (night vision), Traffic monitoring,

   Sports (training and incident analysis)

TARGETED INDUSTRIAL APPLICATIONS

Intelligent Traffic Control
Anti-forging Stamps

Card Counting Systems

Drive Quality Test

Camera Flame Detection

CCTV Fog Penetration

Key Image Search/Index

Security Monitoring

Robust Shadow Detection

Vehicle Segmentation
Visual Tracking Systems

Illegal content (adult) Filter

Scratch Detection

Smart Traffic Monitoring

Vehicle Categorization

Vehicle Wheel alignment

Number Plate Identification

Referrals for Line calls

Different categories of work being done in CV, to solve problems:

2-D image analysis – segmentation, target detection, matching, CBIR;

Pattern Recognition for Objects, scenes;

Feature extraction: Canny, GHT, Snakes, DWT, Corners, SIFT, GLOH, LESH;

Image and Video-based Rendering/Synthesis;

Deepfake detection; Adversarial

3-D multi-camera calibration; Correspondence and stereo; Reconstruction of 3-D Objects and surfaces;

Video and motion analysis; Video analytics; CBVR; Compression;

Multi-sensor data, Decision and feature fusion;

Steganography and Watermarking;

Multimodal AI

The various sub-categories of technology in these related fields are:

- ***Image enhancement,***
- ***Image restoration and filtering,***
- ***Representation and description, Feature extraction,***
- ***Image segmentation,***
- ***Image matching,***
- **Active, Robotic vision**
- **Image synthesis,**
- **Video Analytics**
- ***Image reconstruction***
- ***Range data processing, Stereo image processing***
- ***Computational geometry,***
- ***medical, airborne, underwater***
- ***SAR, LIDAR, VHR-μscopic,***
- ***Neuro-fuzzy techniques,***
- ***DL/ML driven Vision***
- ***Edge computing.***

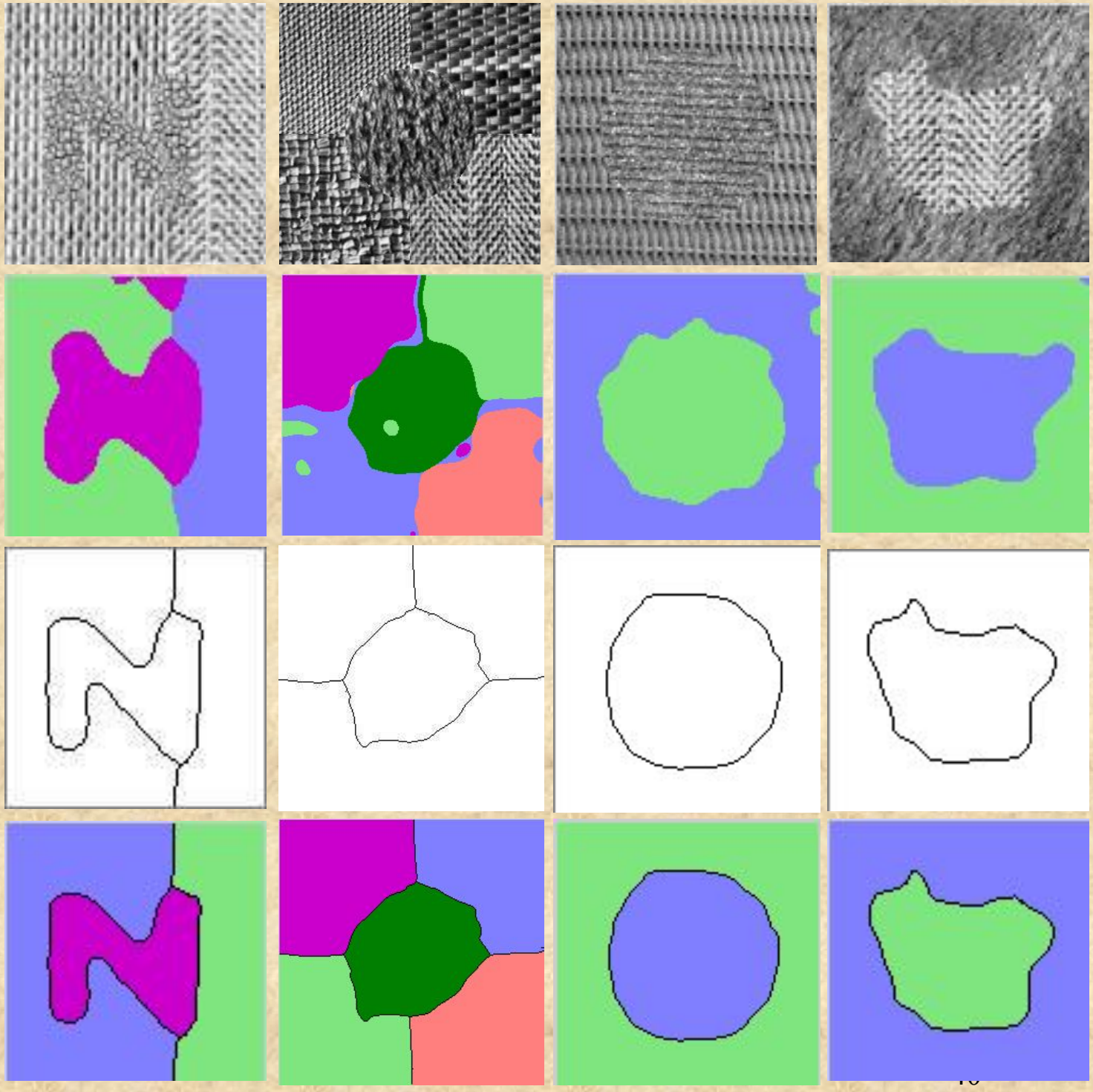
Results of Segmentation

Input Image

Segmented map before integration

Edge map before integration

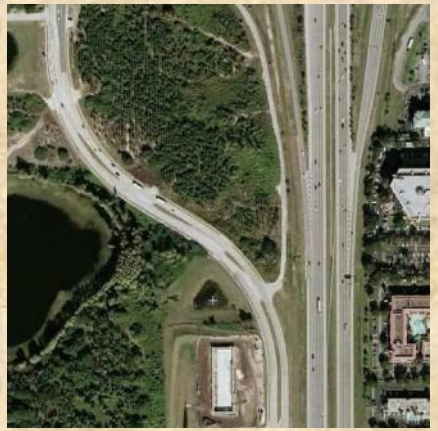
Segmented map and Edge map after integration



Road extraction from Satellite Images

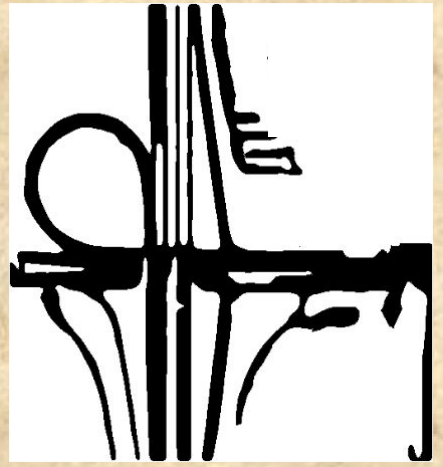
SAT Images



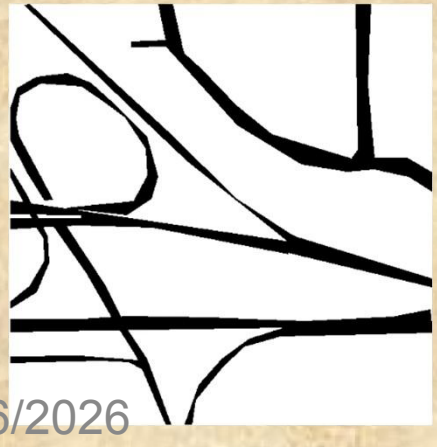
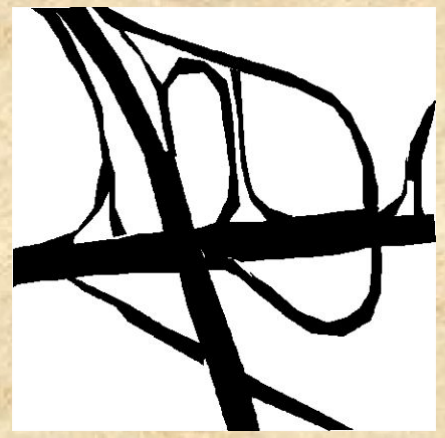


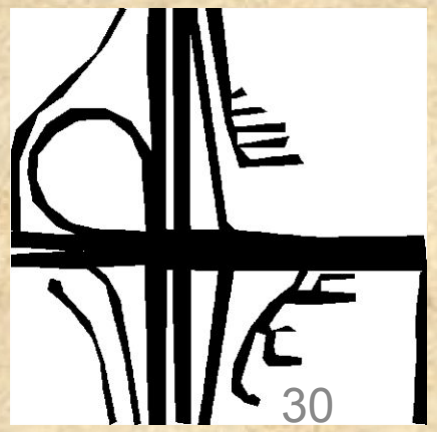

Results



Hand-drawn



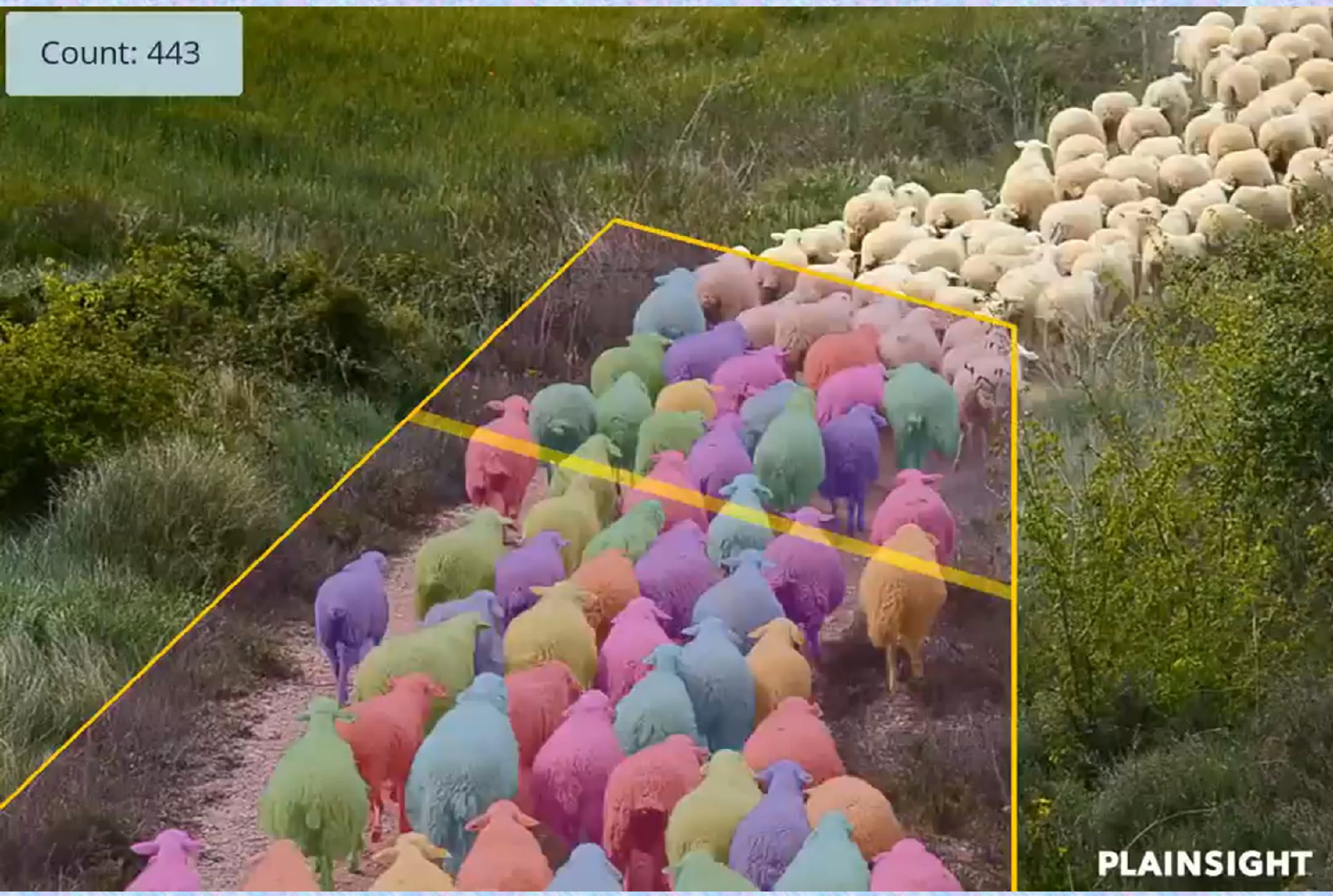
Object Extraction From an Image



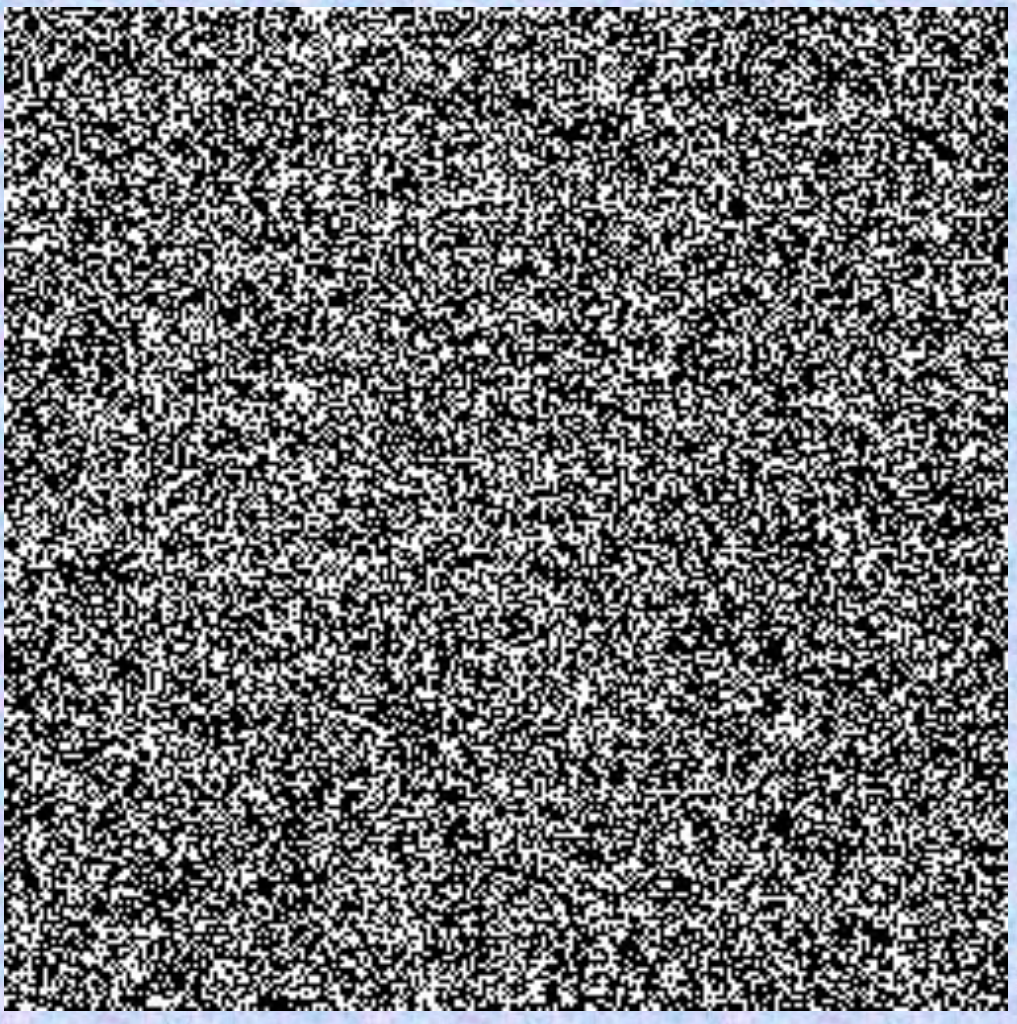








Count: 443
PLAINSIGHT

The Problem Definition



IMRN



IMT

Given a bitmap template (IMT) and a noisy bitmap image IMRN which contains IMT (believe me):

FIND OUT the location of IMT in IMRN !

Go to the next page for more:

Problem explanation for pessimists.

IMR



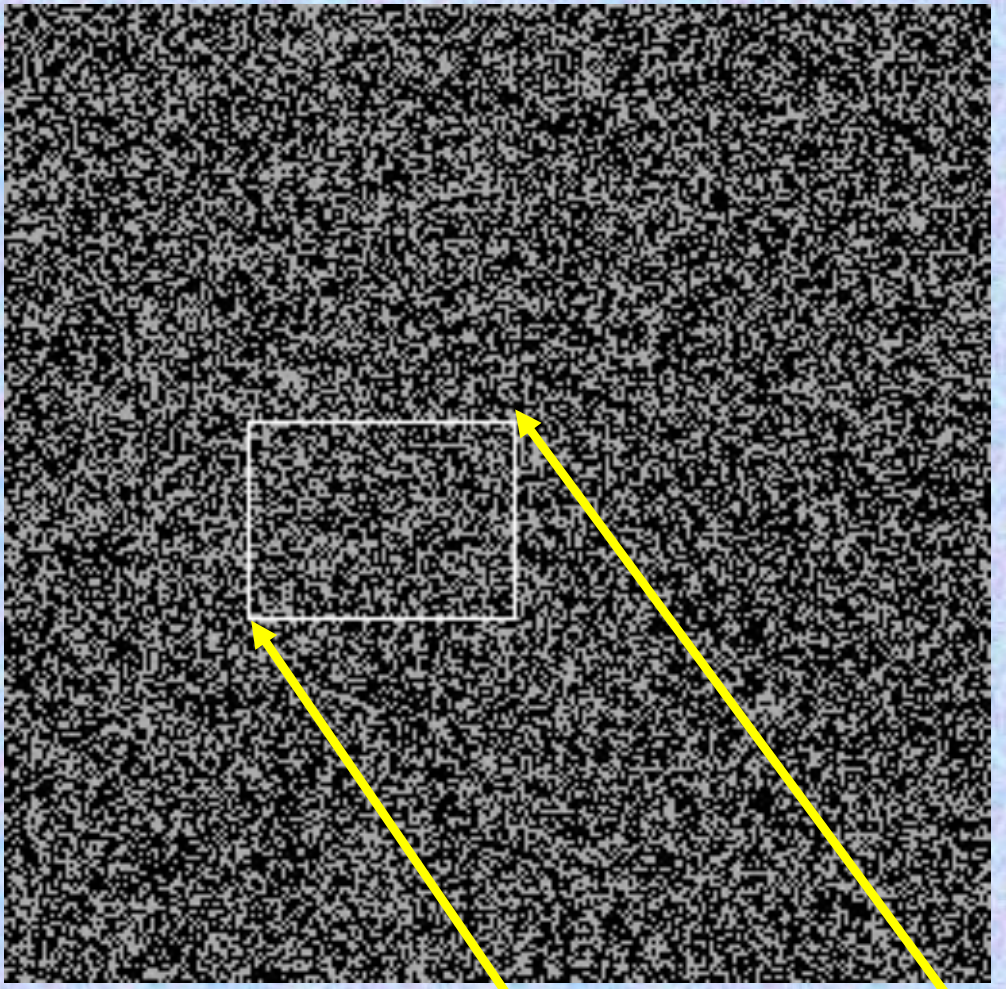
IMT



- IMRN (in previous page) is obtained by adding a large level of "Salt and Pepper" noise onto IMR bitmap image.
- IMT is also obtained from IMR as shown above.

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The RESULT beats the human EYE



IMRN



IMR



IMT

Published almost 3 decades ago
Without GPU and DL

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Thank you